



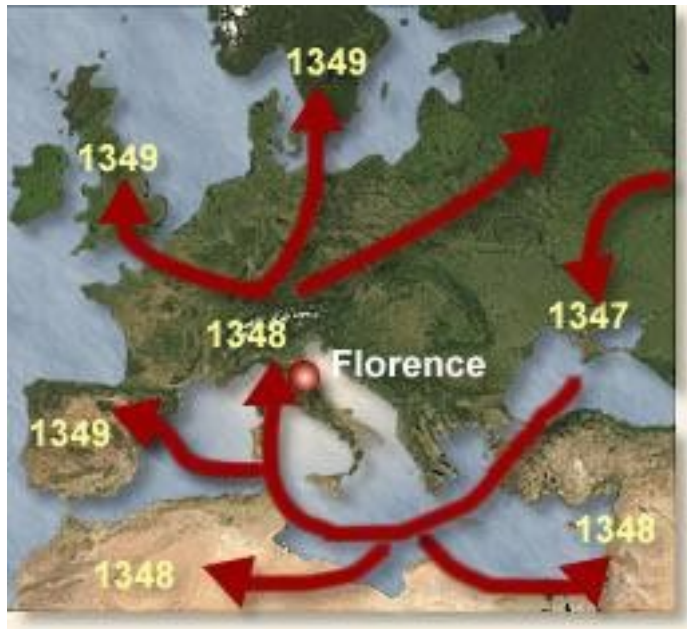
PUBP 233 INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

Week 11: Population Theories

Shivakumar Jolad

THEORIES OF POPULATION

EPIDEMICS AND DEATH



Bubonic Plague- Black Deaths in Europe

Epidemics, Bills of Mortality, Cause of Death

<http://www.eyewitnesstohistory.com/plague.htm>



1346 1347 1348 1349 1350 1351 1352 1353

Approximate border between the Principality of Kiev and the Golden Horde - passage prohibited for Christians.

Land trade routes
Maritime trade routes

Large declines in population
Wiped out 30- 60% of Europe
Population



[http://www.eyewitnesstohistory.com/
plague.htm](http://www.eyewitnesstohistory.com/plague.htm)



[https://www.historytoday.com/archive/black-death-
greatest-catastrophe-ever](https://www.historytoday.com/archive/black-death-greatest-catastrophe-ever)

EPIDEMIC, PANDEMIC, AND ENDEMIC

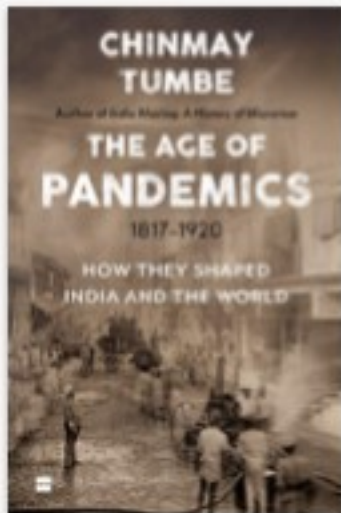
For a disease to become an epidemic, it must suddenly affect many members of a community at the same time. **Time, that is, simultaneity, matters more than space, or geographic coverage.**

For an epidemic to become a pandemic, **both time and space are equally important, that is, the disease has to spread across a sufficiently wide geographical region to be** declared a

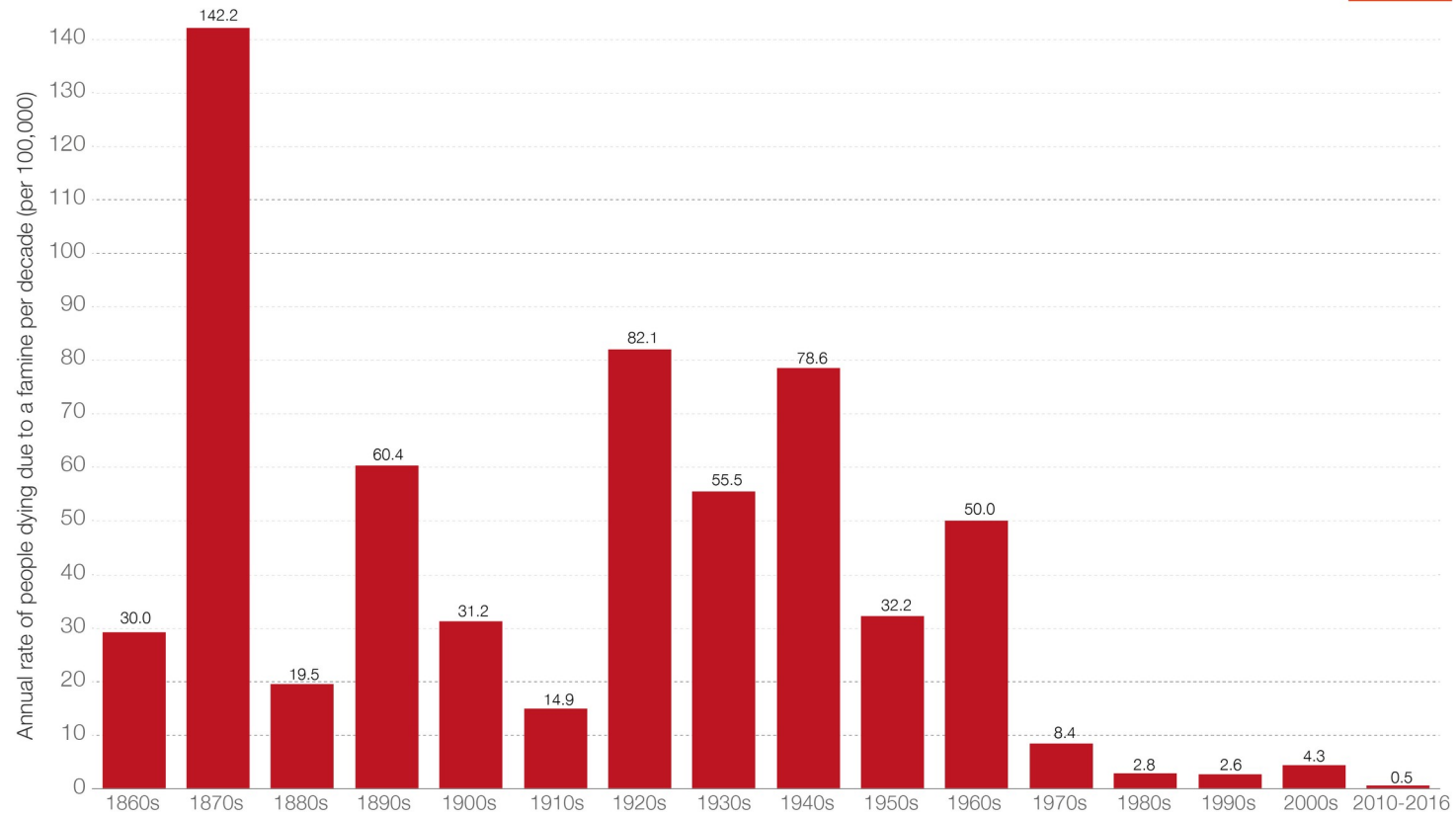
pandemic
Maari and

Mahamaari

Endemic
disease?

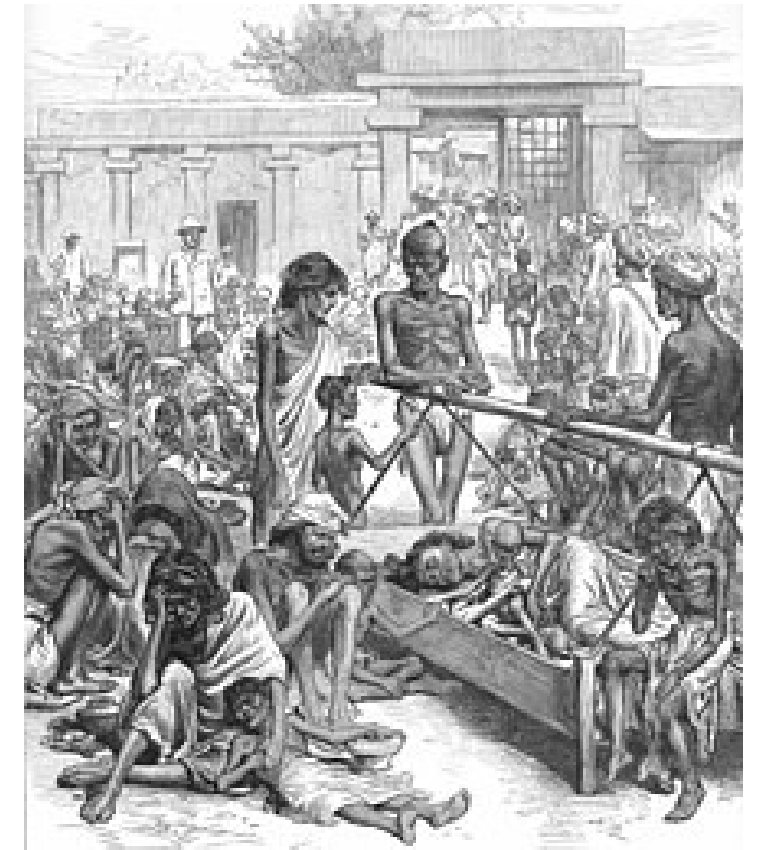


Annual rate of people dying due to a famine globally, per decade



Data source: The rate of the excess mortality due to famines shown here is presented in detail on OurWorldinData.org/famines [The dataset was constructed by Joe Hasell and Max Roser].
How the rate is calculated: The annual rate is calculated for each decade by dividing the total number of famine victims in each decade by the average size of the world population in that decade and dividing the resulting ratio by 10 (and for the data from 2010 to 2016 it is divided by 7) to arrive at a yearly rate. To express it as the rate per 100,000 people, this annual rate is then multiplied by 100,000.
For famines that happened at the end of a decade and the beginning of the next decade the famine victims are split by decade on a year by year basis.
For famines for which different excess mortality estimates are published the midpoint between these estimates was chosen here.
This visualization is available at OurWorldinData.org. There you find the full dataset and more research and visualizations on famines and global development.

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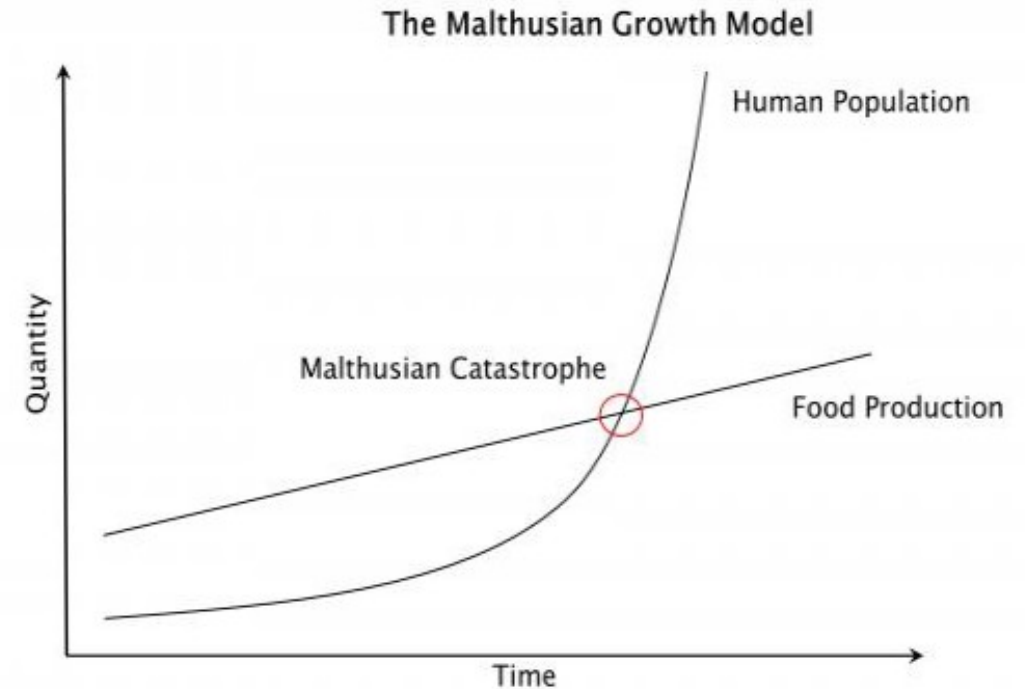
MALTHUSIAN THEORY

“The power of population is so superior to the power of the earth to produce subsistence for man, that premature death must in some shape or other visit the human race. .. Should success be still incomplete, gigantic inevitable famine stalks in the rear, and with one mighty blow levels the population with the food of the world”.



Thomas Malthus (1766-1834)

Photo from wiki :
http://en.wikipedia.org/wiki/Thomas_Robert_Malthus



An Essay on the Principle of Population

*An Essay on the Principle of Population, as it
Affects the Future Improvement of Society
with Remarks on the Speculations of Mr. Godwin,
M. Condorcet, and Other Writers.*

Thomas Malthus

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PREFACE..... VII

CHAPTER 11

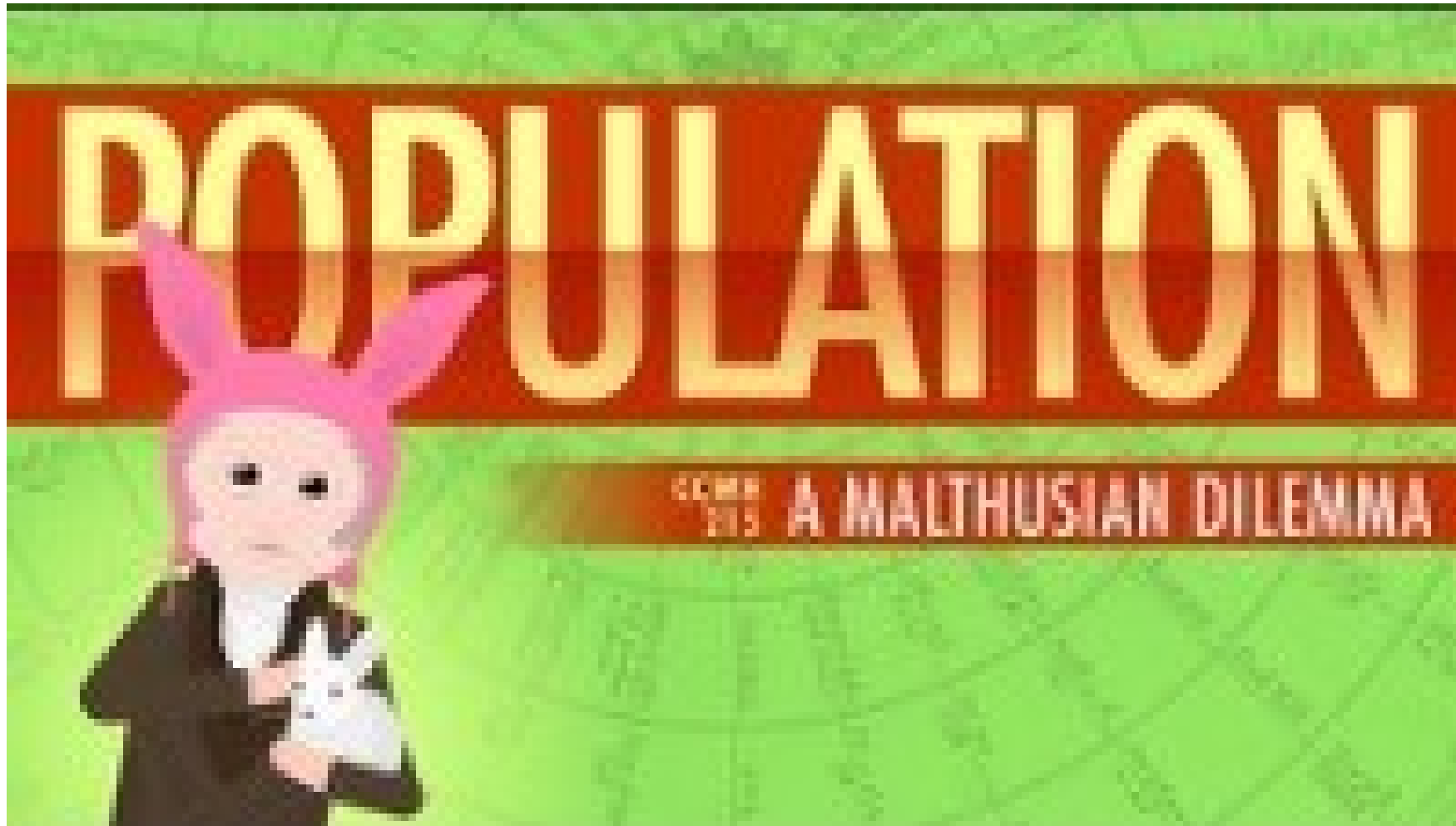
Question stated - Little prospect of a determination of it, from the enmity of the opposing parties - The principal argument against the perfectibility of man and of society has never been fairly answered - Nature of the difficulty arising from population - Outline of the principal argument of the Essay

CHAPTER 26

The different ratio in which population and food increase - The necessary effects of these different ratios of increase - Oscillation produced by them in the condition of the lower classes of society - Reasons why this oscillation has not been so much observed as might be expected - Three propositions on which the general argument of the Essay depends - The different states in which mankind have been known to exist proposed to be examined with reference to these three propositions.

CHAPTER 312

The savage or hunter state shortly reviewed - The shepherd state, or the tribes of barbarians that overran the Roman Empire - The superiority of the power of population to the means of subsistence - the cause of the great tide of Northern Emigration.



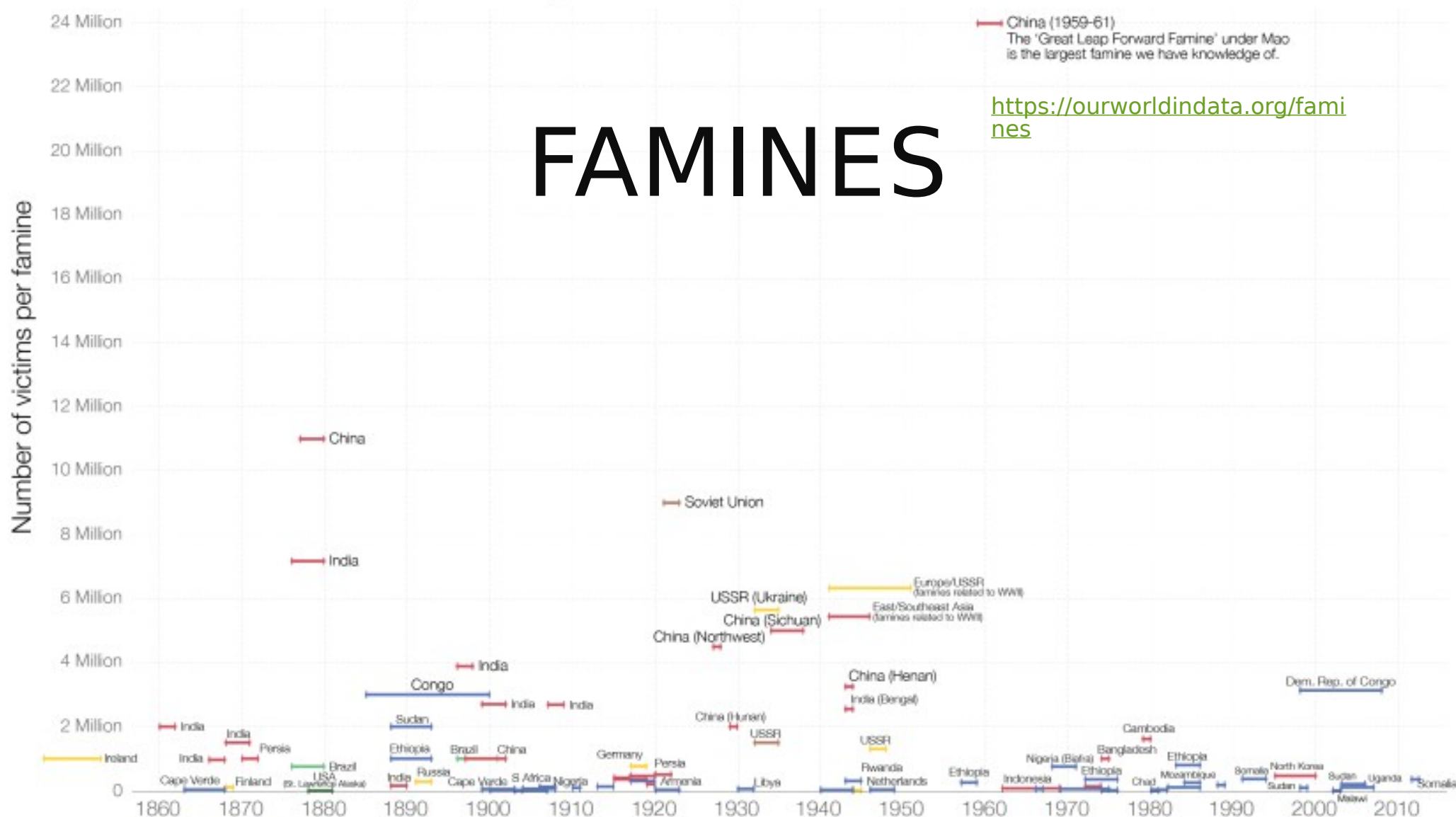
From:
https://www.youtube.com/watch?v=QAKW_i0bDpQ 1:20 s

The number of victims in famines worldwide, from 1860 to 2016

The excess mortality due to famines shown here is presented in detail on [OurWorldInData.org/famines](https://ourworldindata.org/famines).

The length of the line shows the duration of the famine, the position of the line shows the total number of victims for each famine, the color of the line shows the continent where the famine happened.

For famines for which different mortality estimates are published the midpoint between these estimates is shown here.



Data source: [OurWorldinData.org/famines](https://ourworldindata.org/famines) [The dataset was constructed by Joe Hasell and Max Roser]

This visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find the full dataset and more research on famines and global development.

Licensed under CC-BY-SA by the author Max Roser.

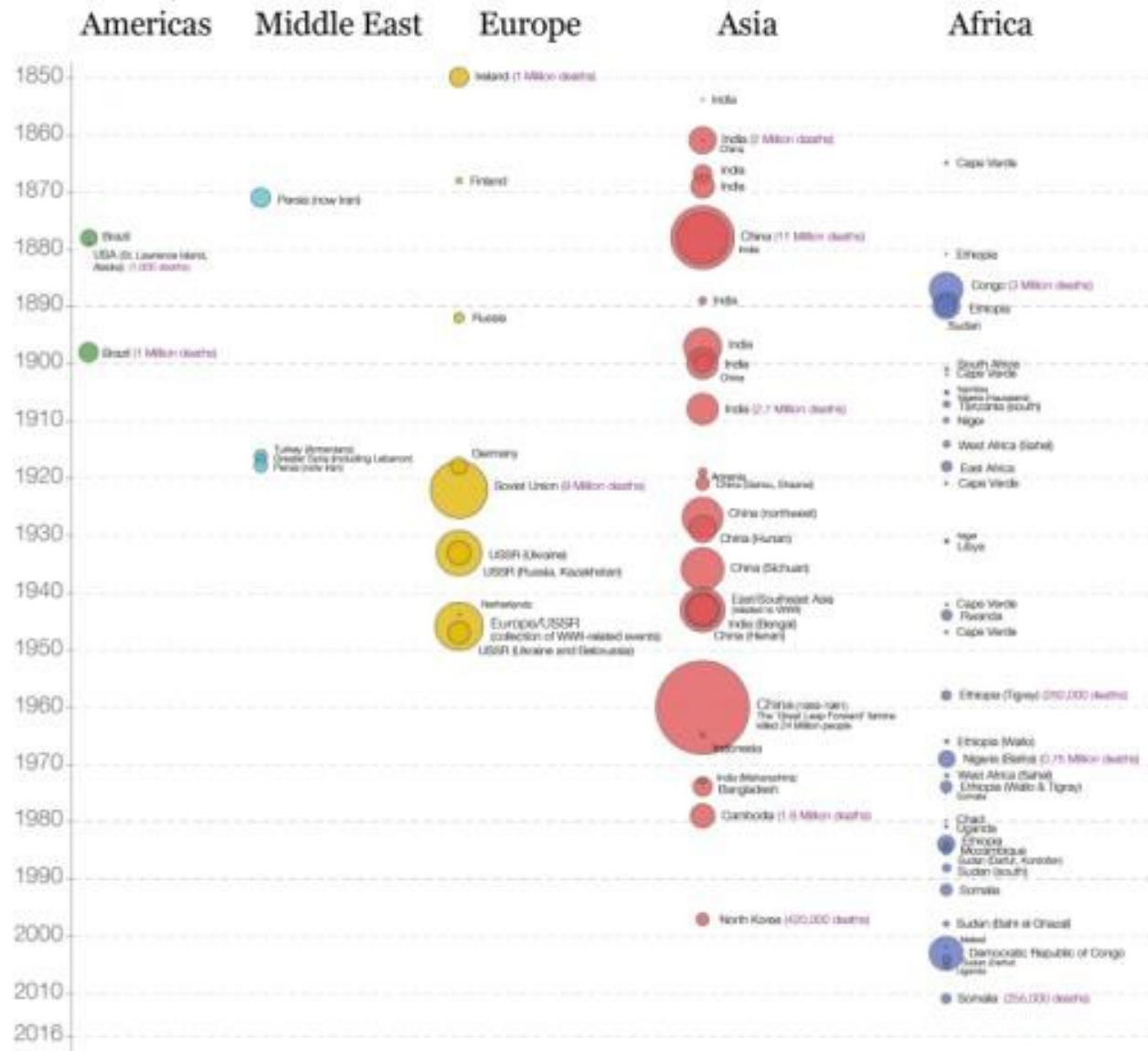
Famines by world region, 1860-2016

Our World
in Data

The size of the bubble represents the death count of the famine (excess mortality).

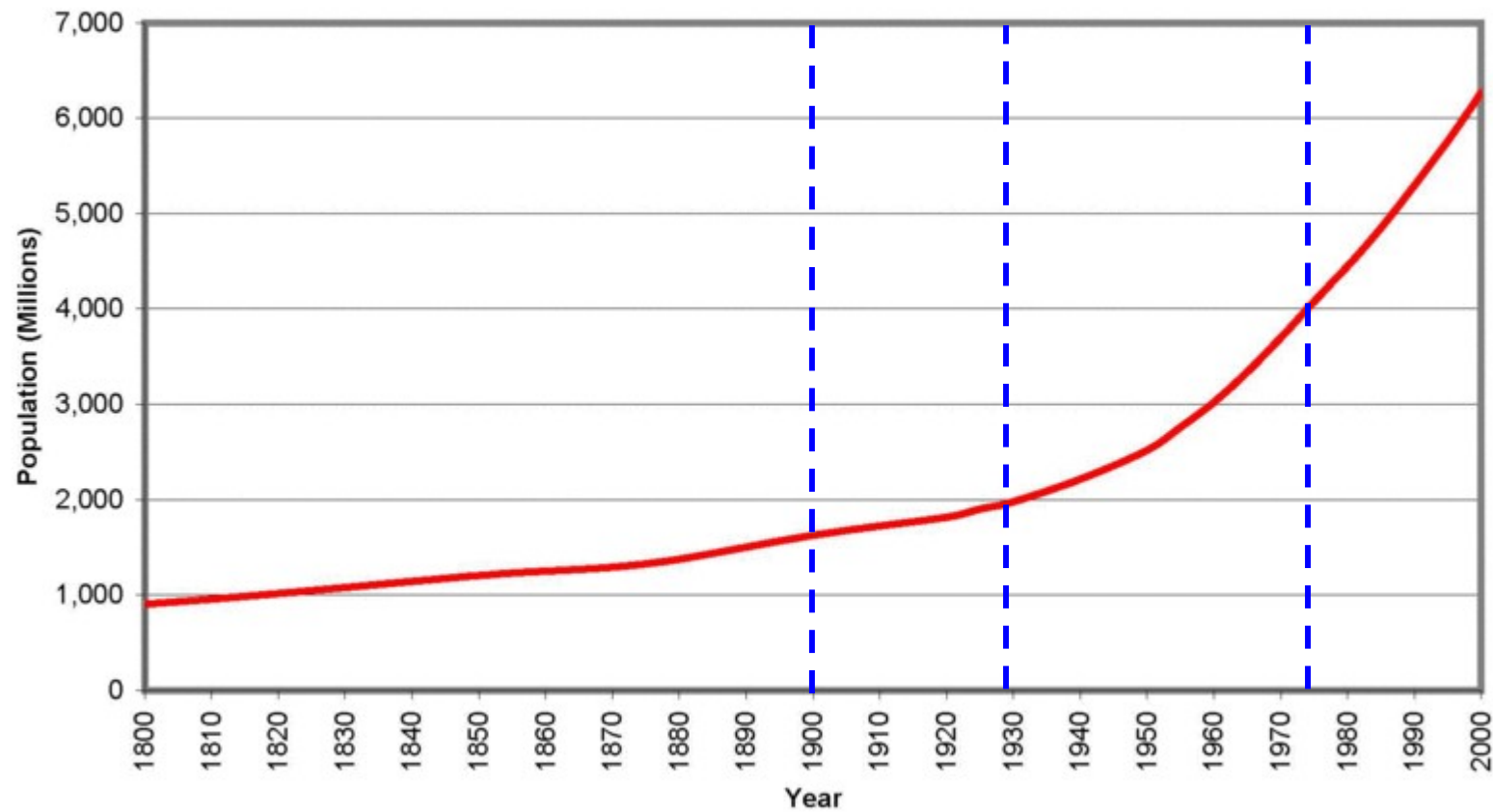
For famines for which different excess mortality estimates are published the midpoint between these estimates was chosen here.

Detailed information on this dataset is available at [OurWorldInData.org/famines](https://ourworldindata.org/famines).



<https://ourworldindata.org/famines>

World Population, 1800 - 2000

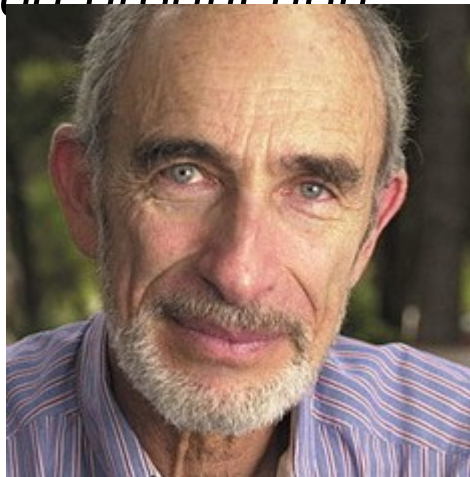


Source: J. Bradford DeLong, "Estimating World GDP, One Million B.C. - Present." http://www.j-bradford-delong.net/TCEH/1998_Draft/World_GDP/Estimating_World_GDP.html. Accessed Mar 5, 2008.

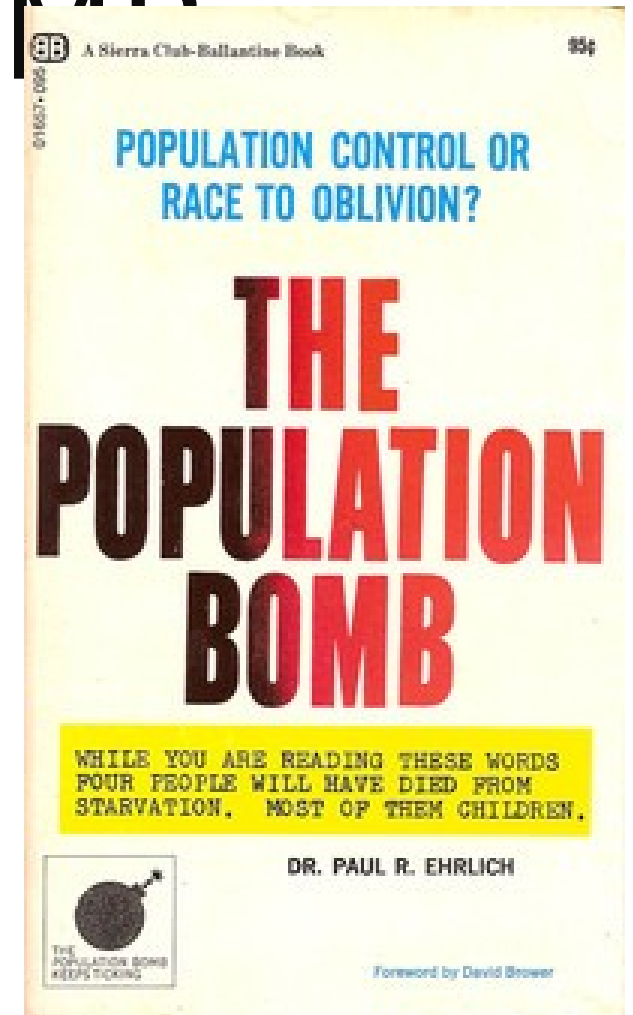
© 2008. Michael W. Kruse

THE POPULATION BOMB

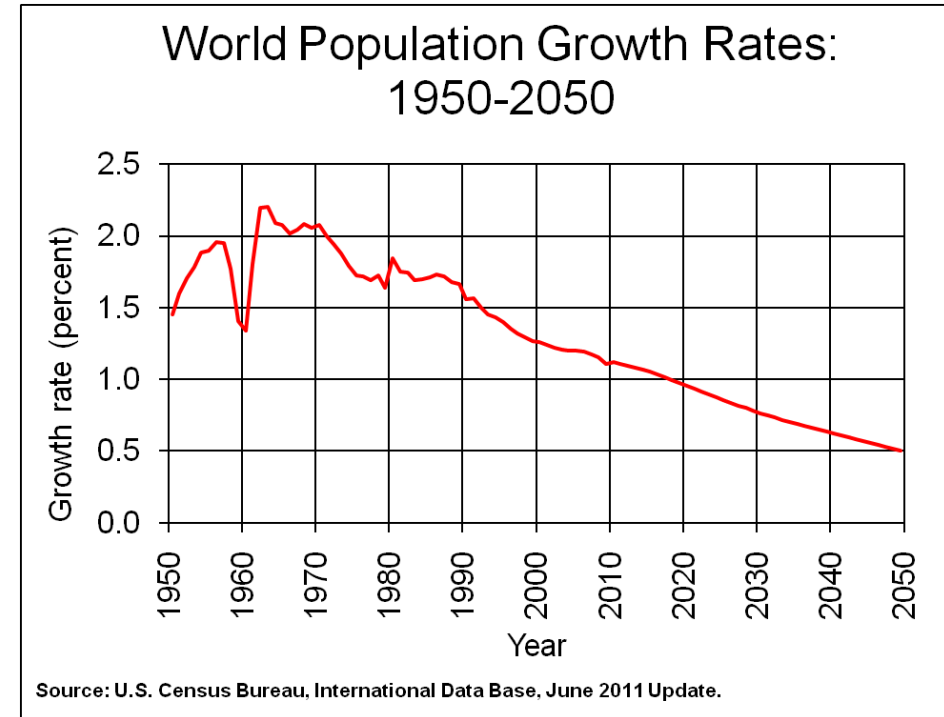
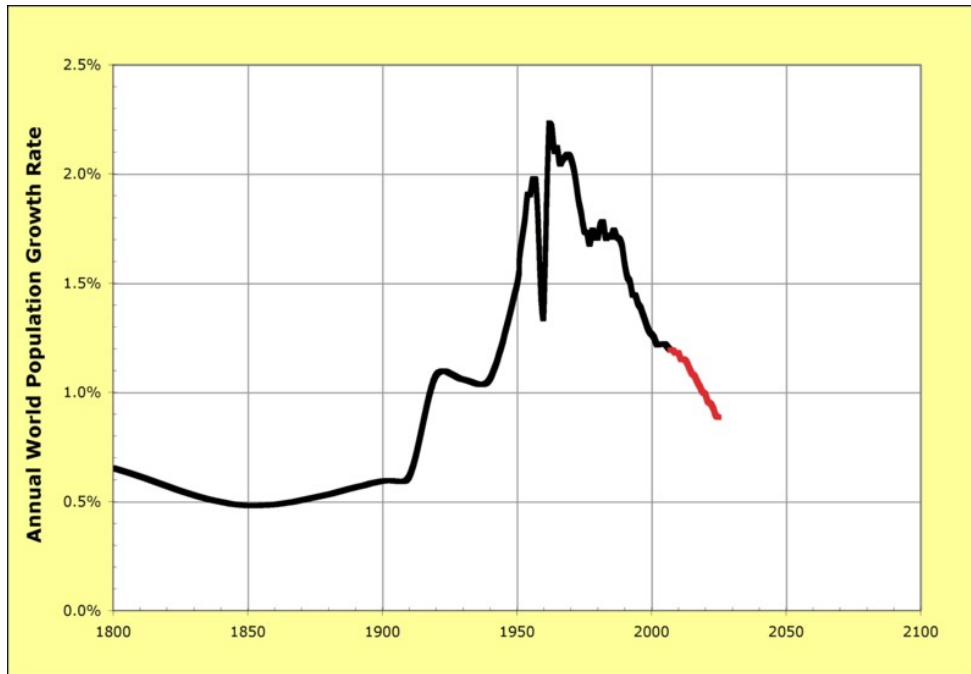
"The battle to feed all of humanity is over. In the 1970s hundreds of millions of people will starve to death in spite of any crash programs embarked upon now. At this late date nothing can prevent a substantial increase in the world death rate."
"We must rapidly bring the world population under control, reducing the growth rate to zero or making it negative. Conscious regulation of human numbers must be achieved. Simultaneously we must, at least temporarily, greatly increase our food production."



**Paul Ralph
Ehrlich, 1968**



WORLD POPULATION GROWTH RATE



REALITY CHECK- LIFE EXPECTANCY

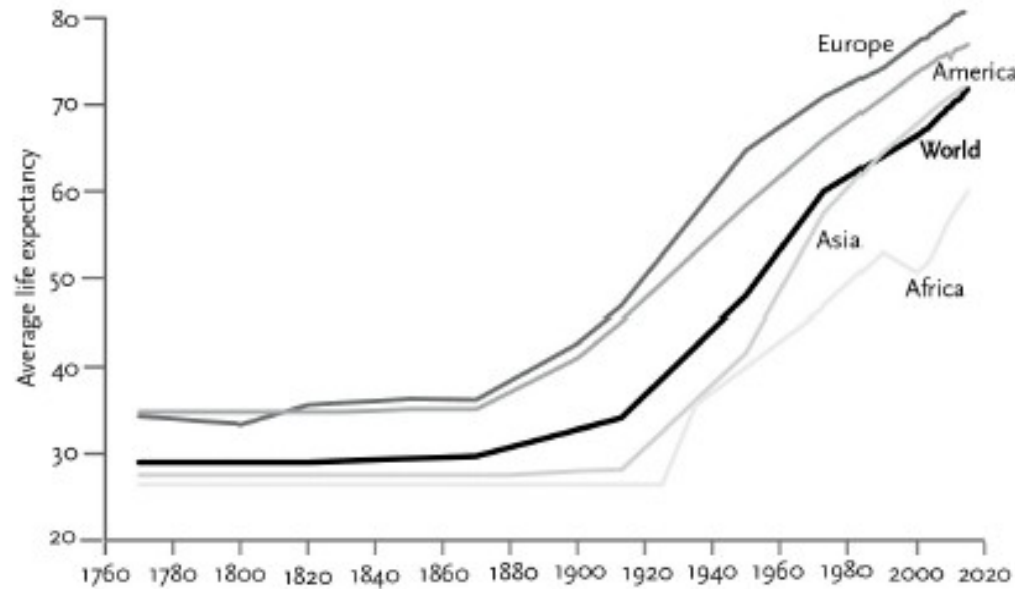


Figure 5-1: Life expectancy, 1771–2015

Sources: *Our World in Data*, Roser 2016n, based on data from Riley 2005 for the years before 2000 and from the World Health Organization and the World Bank for the subsequent years. Updated with data provided by Max Roser.

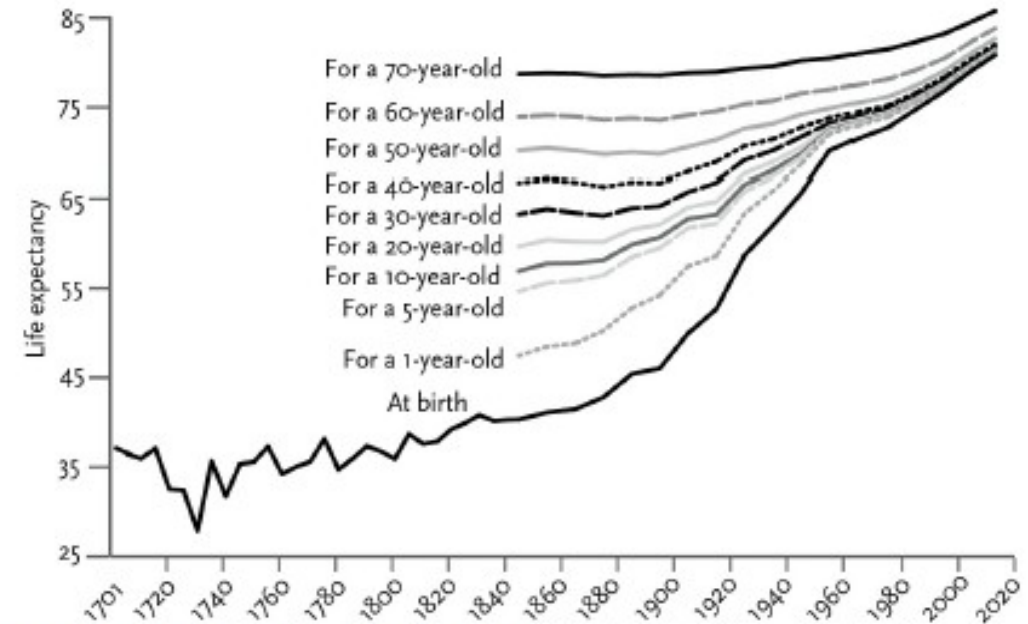


Figure 5-4: Life expectancy, UK, 1701–2013

Sources: *Our World in Data*, Roser 2016n. Data before 1845 are for England and Wales and come from OECD Clío Infra, van Zanden et al. 2014. Data from 1845 on are for mid-decade years only, and come from the *Human Mortality Database*, <http://www.mortality.org/>.



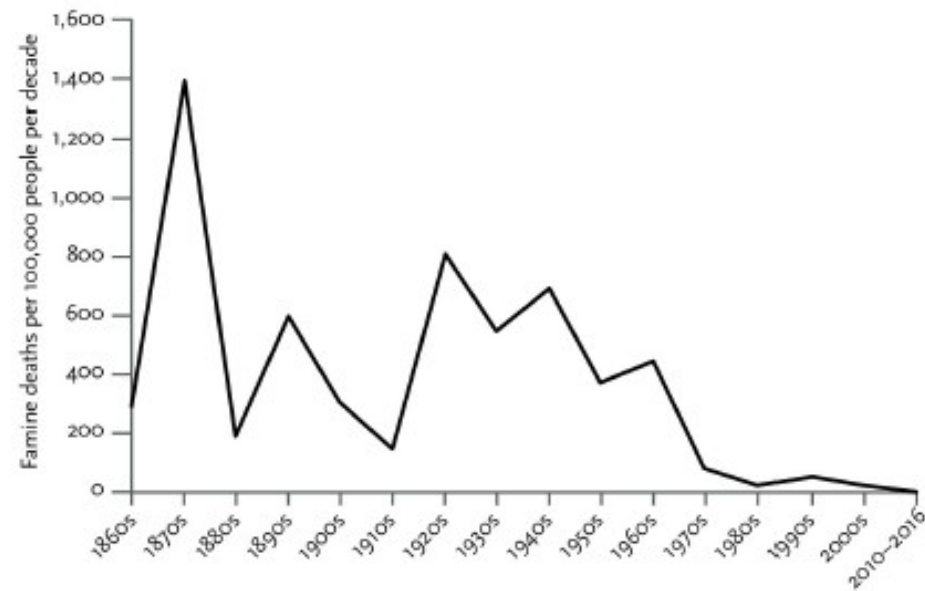


Figure 7-4: Famine deaths, 1860–2016

Sources: *Our World in Data*, Hasell & Roser 2017, based on data from Devereux 2000; Ó Gráda 2009; White 2011, and EM-DAT, *The International Disaster Database*, <http://www.emdat.be/>; and other sources. "Famine" is defined as in Ó Gráda 2009.

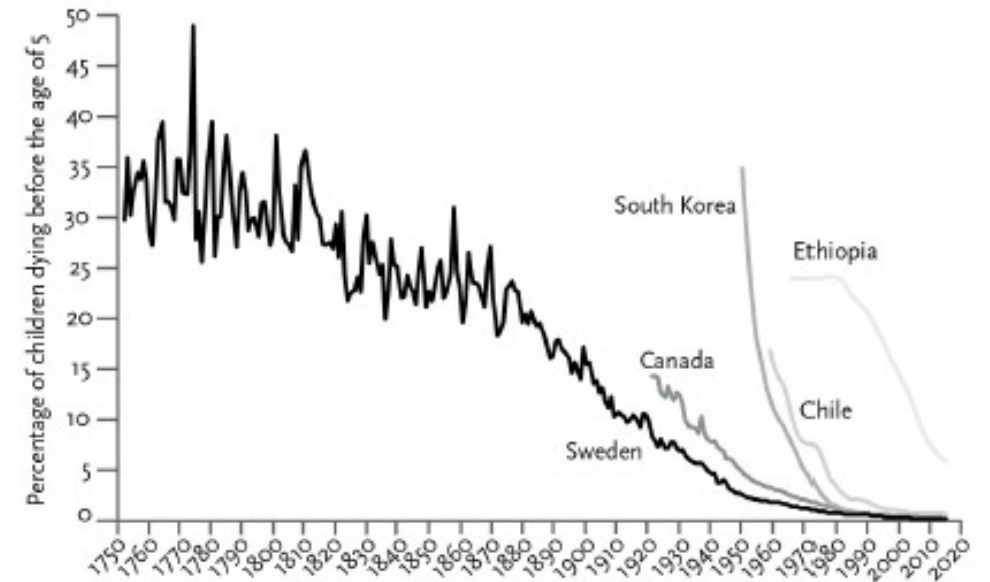


Figure 5-2: Child mortality, 1751–2013

Sources: *Our World in Data*, Roser 2016a, based on data from the UN Child Mortality estimates, <http://www.childmortality.org/>, and the *Human Mortality Database*, <http://www.mortality.org/>.



CALORIFIC INTAKE AND UNDERNOURISHMENT

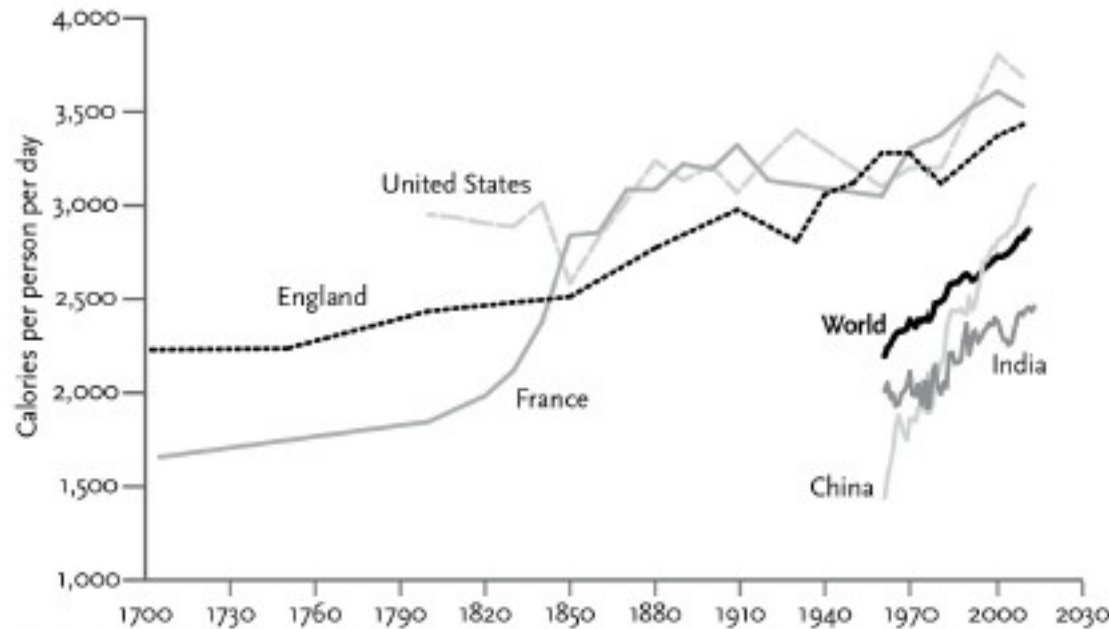


Figure 7-1: Calories, 1700–2013

Sources: United States, England, and France: *Our World in Data*, Roser 2016d, based on data from Fogel 2004. China, India, and the World: Food and Agriculture Organization of the United Nations, <http://www.fao.org/faostat/en/#data>.

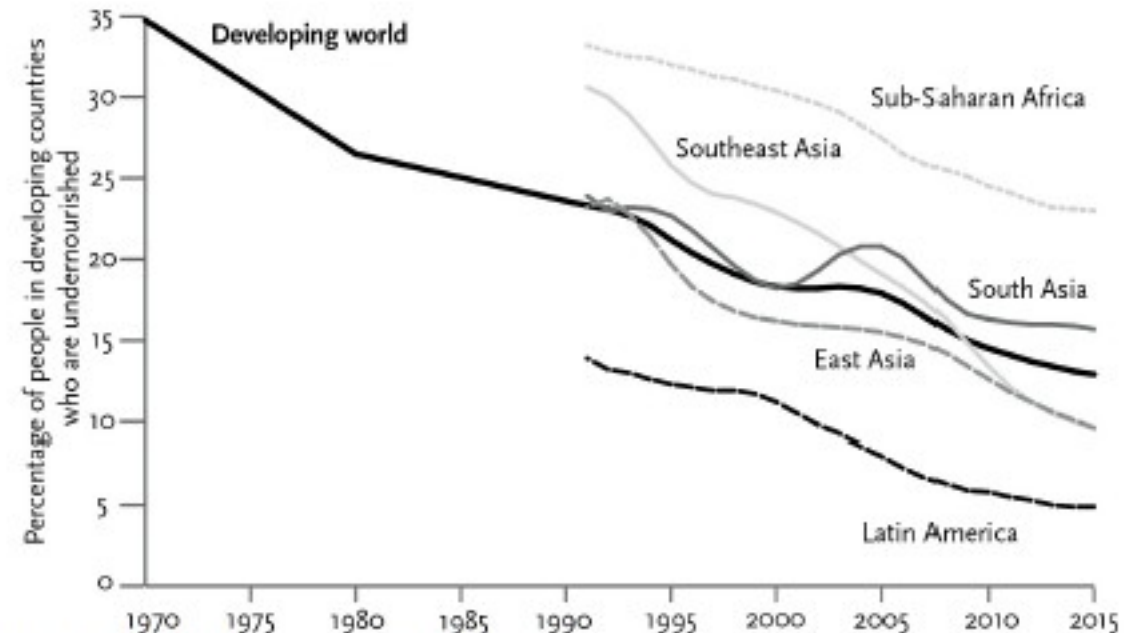


Figure 7-3: Undernourishment, 1970–2015

Source: *Our World in Data*, Roser 2016j, based on data from the Food and Agriculture Organization 2014, also reported in <http://www.fao.org/economic/ess/ess-fs/ess-fadata/en/>.

**BREAK KE BAAD (POST
FERTILITY)**

KARL MARX - THEORY

Opposed Malthusian View; Population growth, not related to the alleged ignorance or moral inferiority of the poor, **but is a consequence of the capitalist economic system.**

Machinery → unemployment → reserve army of labour → wages go down → Poverty increase

Poverty, hunger and other social ills are the result of socially unjust practices associated with capitalism.

Landlordism, unfavorable and high man-land ratio, uncertainty regarding land tenure system and the like are responsible for low food production in a country.

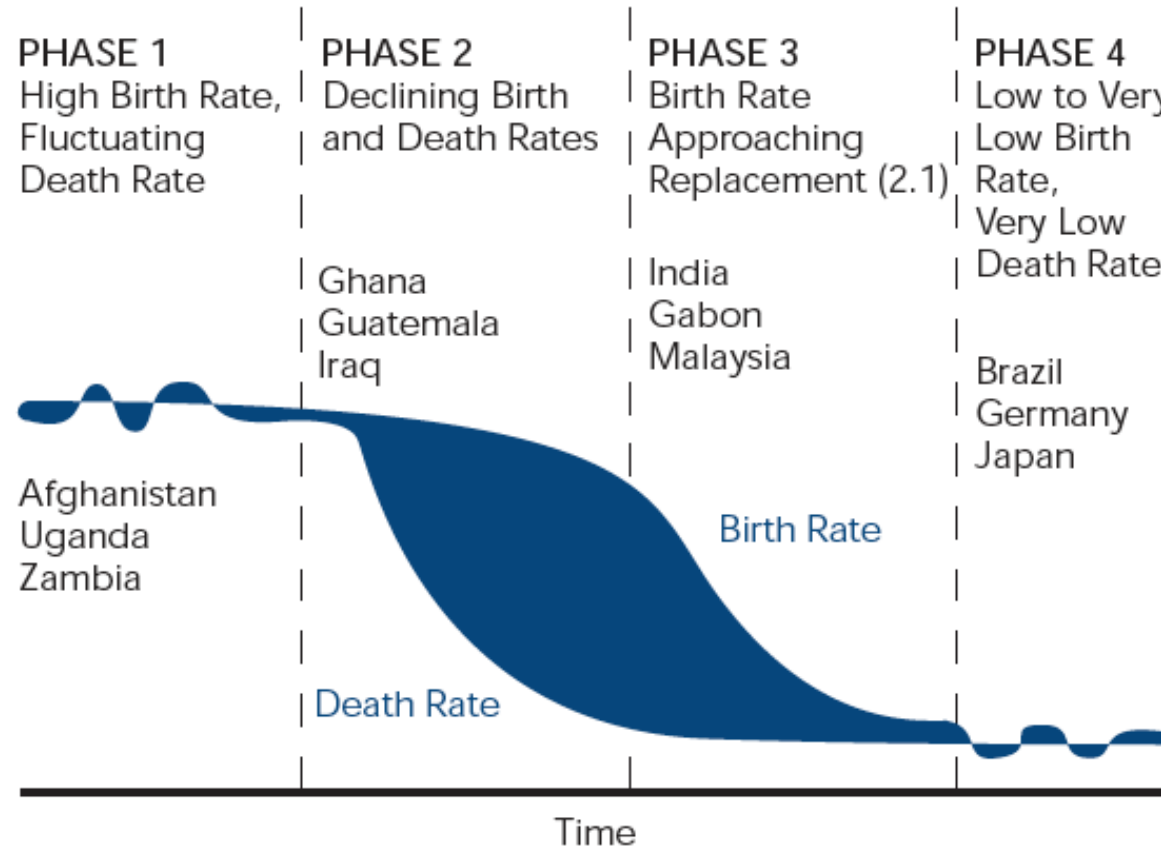
Only in places where the production of food is not adequate does population growth become a problem.



Karl Marx (1818-1883)

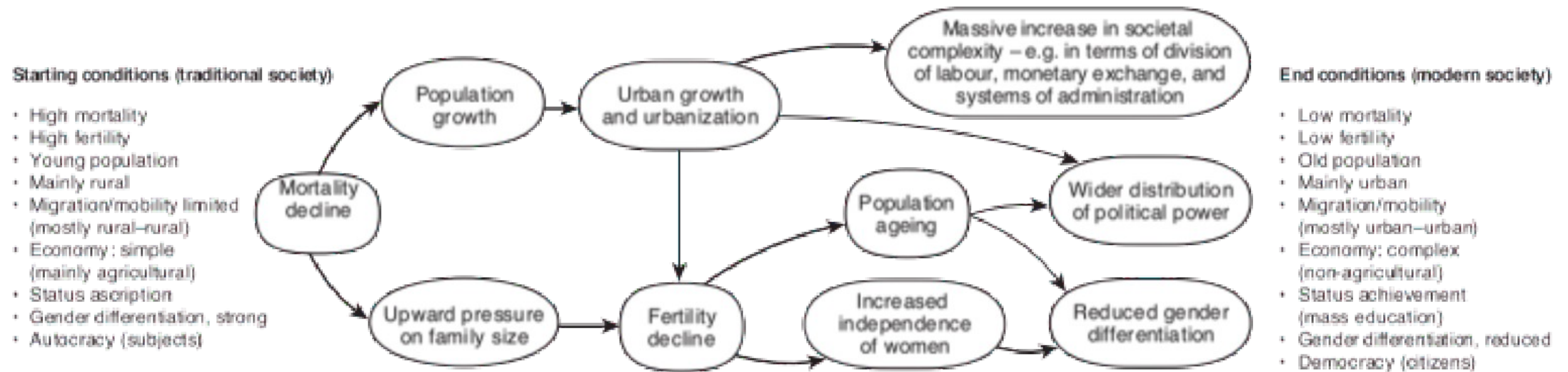
DEMOGRAPHIC TRANSITIONS

The Classic Phases of Demographic Transition



Frank W. Notestein (1902-1983)

DEMOGRAPHIC TRANSITION

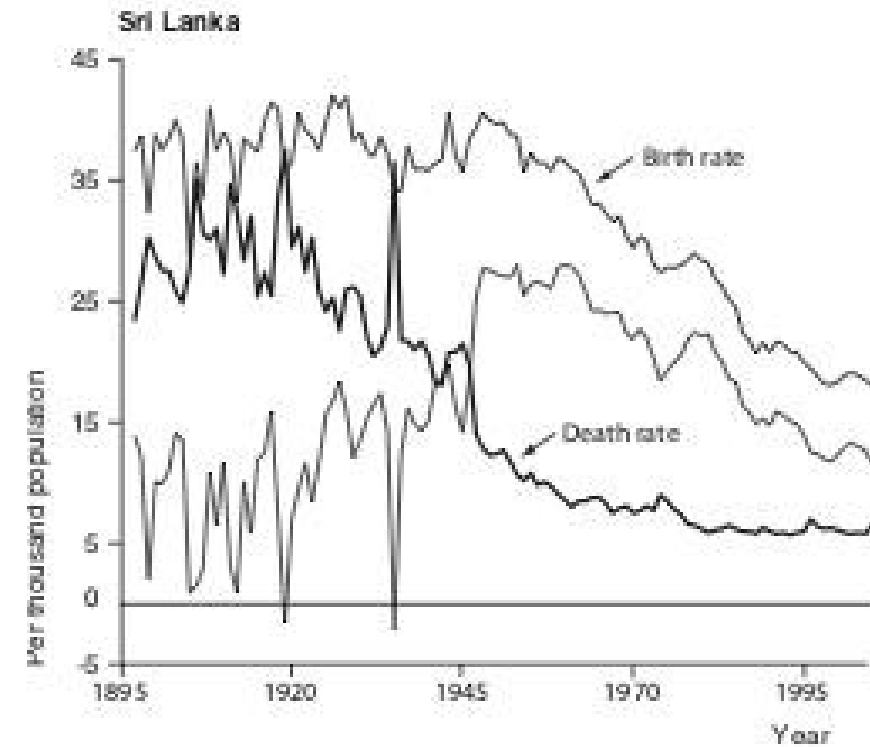
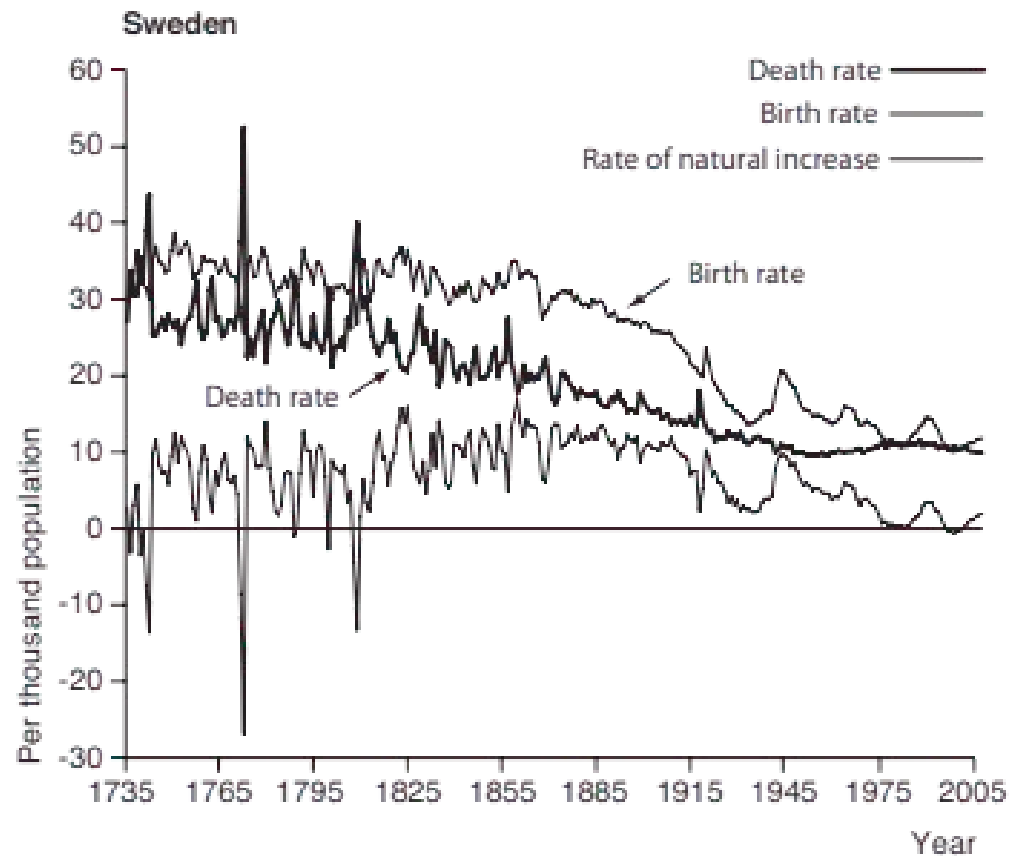


Notes: Clearly, the diagram is highly simplified and selective. For example, it makes little reference to the social, economic and educational consequences of the transition. There is no attempt to specify interdependent processes and feedback effects. More important causal relationships are indicated in bold. Later in the book, Figure 5.3 provides elaboration on how mortality decline leads to urban growth and urbanization; and Table 6.1 provides some elaboration regarding the huge increase in societal complexity that results from the expansion of the urban sector. See also Dyson (2001).

Figure 2.1 The role of the demographic transition in world development—key relationships

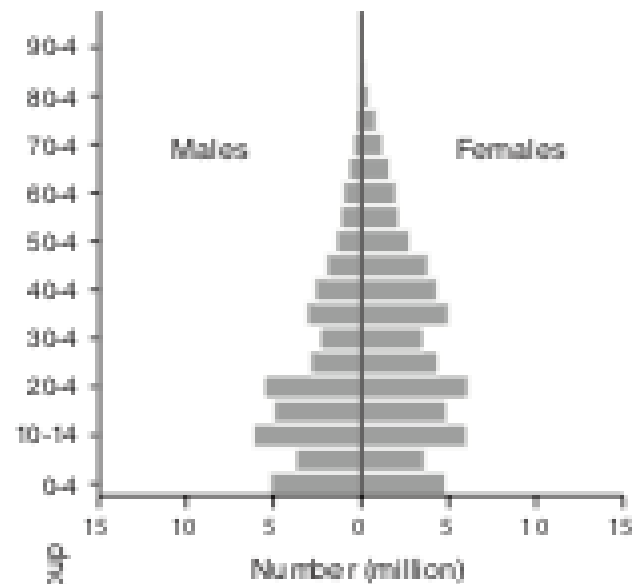
Population and development
the demographic transition

Tim Dyson

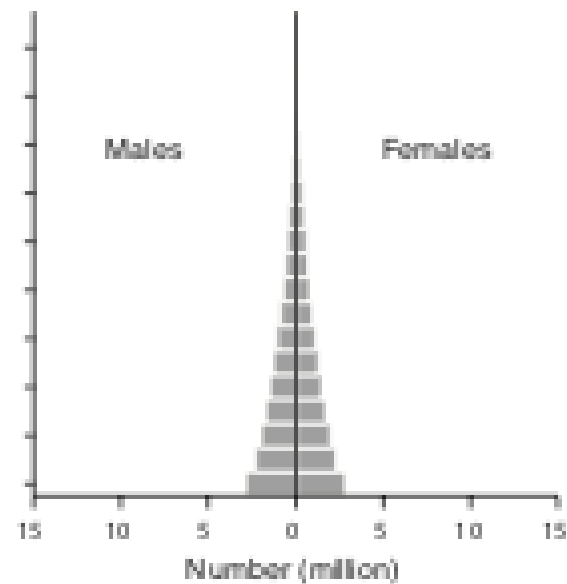


Principal sources: Sweden 1735–1984 (Chesnais 1992), 1985–2008 (Statistics Sweden 2009); Sri Lanka 1897–1934 (Mitchell 1982), 1935–2005 (Sri Lanka 2009).

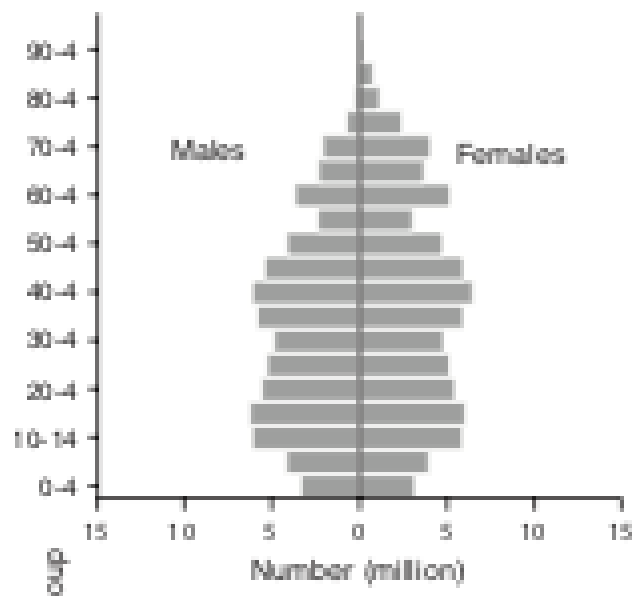
Russia 1950 (Population: 102.7 million)



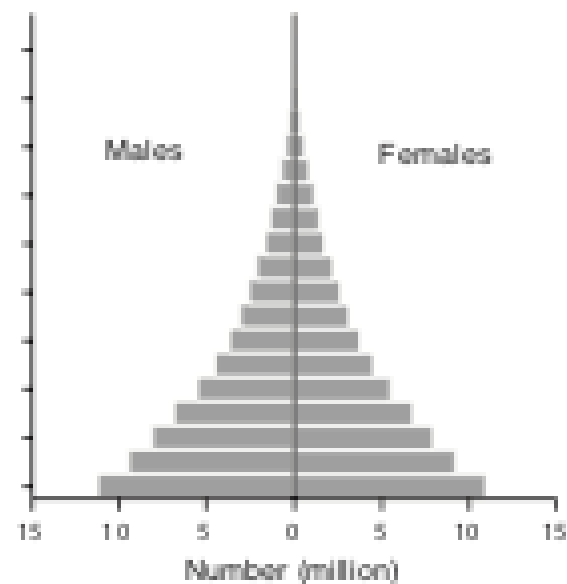
Nigeria 1950 (Population: 36.7 million)



Russia 2000 (Population: 146.7 million)



Nigeria 2000 (Population: 124.8 million)



PHASE I- PRE TRANSITION

pre-transitional Europe was characterised in general by high and fluctuating mortality, and moderately high fertility.

Life expectancy ranged between 25 and 40 years, and experienced large fluctuations in relatively short periods of time.

Population size was often reduced by "mortality crises", caused by outbreaks of plague or other infectious disease. On a local scale, crop failure and famine played a role.

Total fertility rates were moderately high, ranging between 4 and just under 6 children per woman on average.

Fertility rates were determined largely by age at marriage and proportion of the population remaining celibate. These in turn were influenced by real wages, as higher wages meant couples could afford to marry and form a household earlier than when wages were low.

Population growth was slow, due to high mortality and moderate fertility.

PHASE II- EARLY TRANSITION

In the 18th century Western Europe saw a reduction in the magnitude and frequency of mortality crises

Over the course of the 19th century, life expectancy at birth rose from around 40 years to over 50 years by the first decade of the 20th century.

Initially there was little improvement in infant mortality due to terrible care and feeding of infants. **Decline in mortality affected young adults primarily.**

Improvements in infant mortality only began from the beginning of the 20th century.

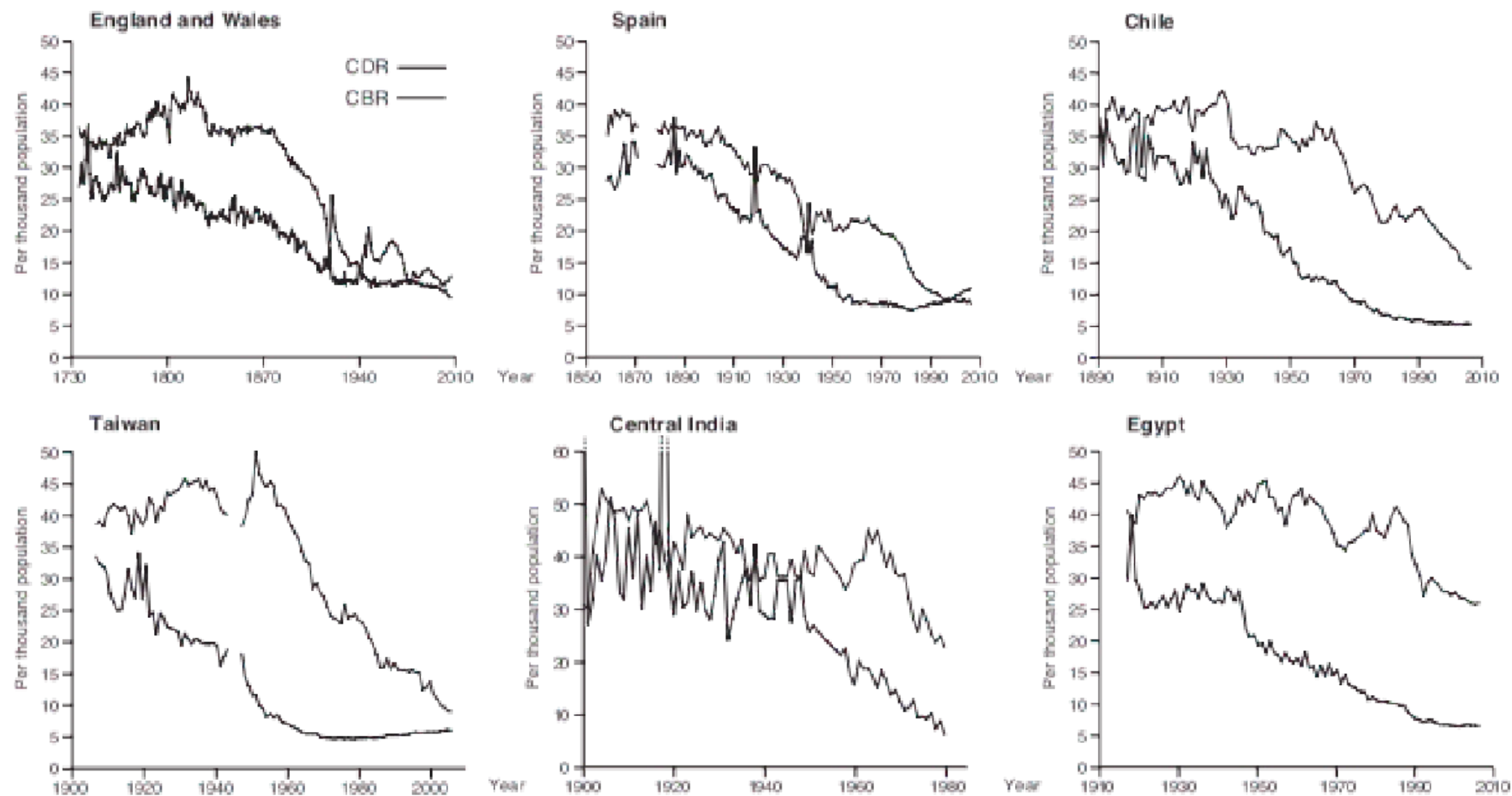
http://papp.iussp.org/sessions/pap_p101_s01

LATE TRANSITION

Late Transition: Fertility begins to decline (having lagged behind the decline in mortality).

The extent to which historical transitions followed this pattern varies by country.

- Transition in England and Wales did follow the classical pattern, **in France fertility and mortality declined simultaneously from the mid-18th century.**



Principal sources: England and Wales (Wrigley and Schofield 1981; Chesnais 1992; ONS 2009); Spain (Chesnais 1992); Chile (Mitchell 1983; INE-CELADE 2000); Taiwan (Mitchell 1982; Republic of China 2007); Central India (Dyson 1989); Egypt (Chesnais 1992). Rates were also taken from United Nations *Demographic Yearbook* (various years).

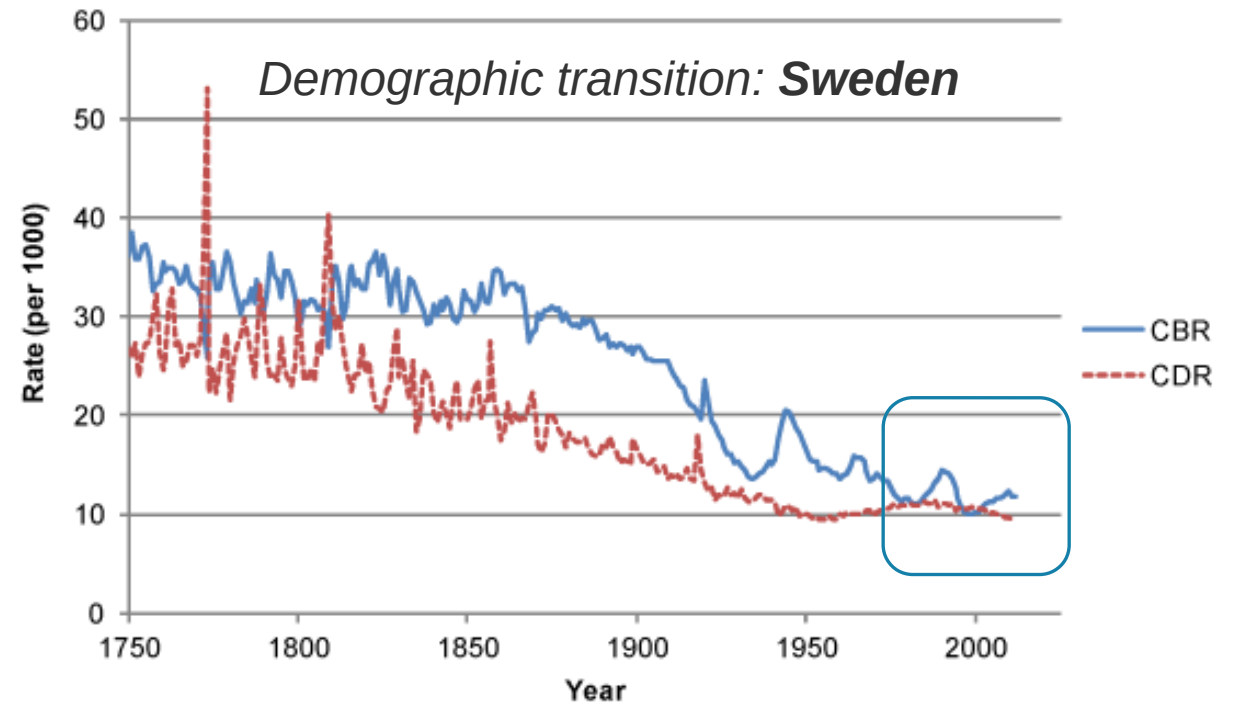
Figure 4.1 The demographic transition—vital rates in six populations

POST TRANSITION

Low mortality
concentrated in the
oldest ages,

low fertility rates,

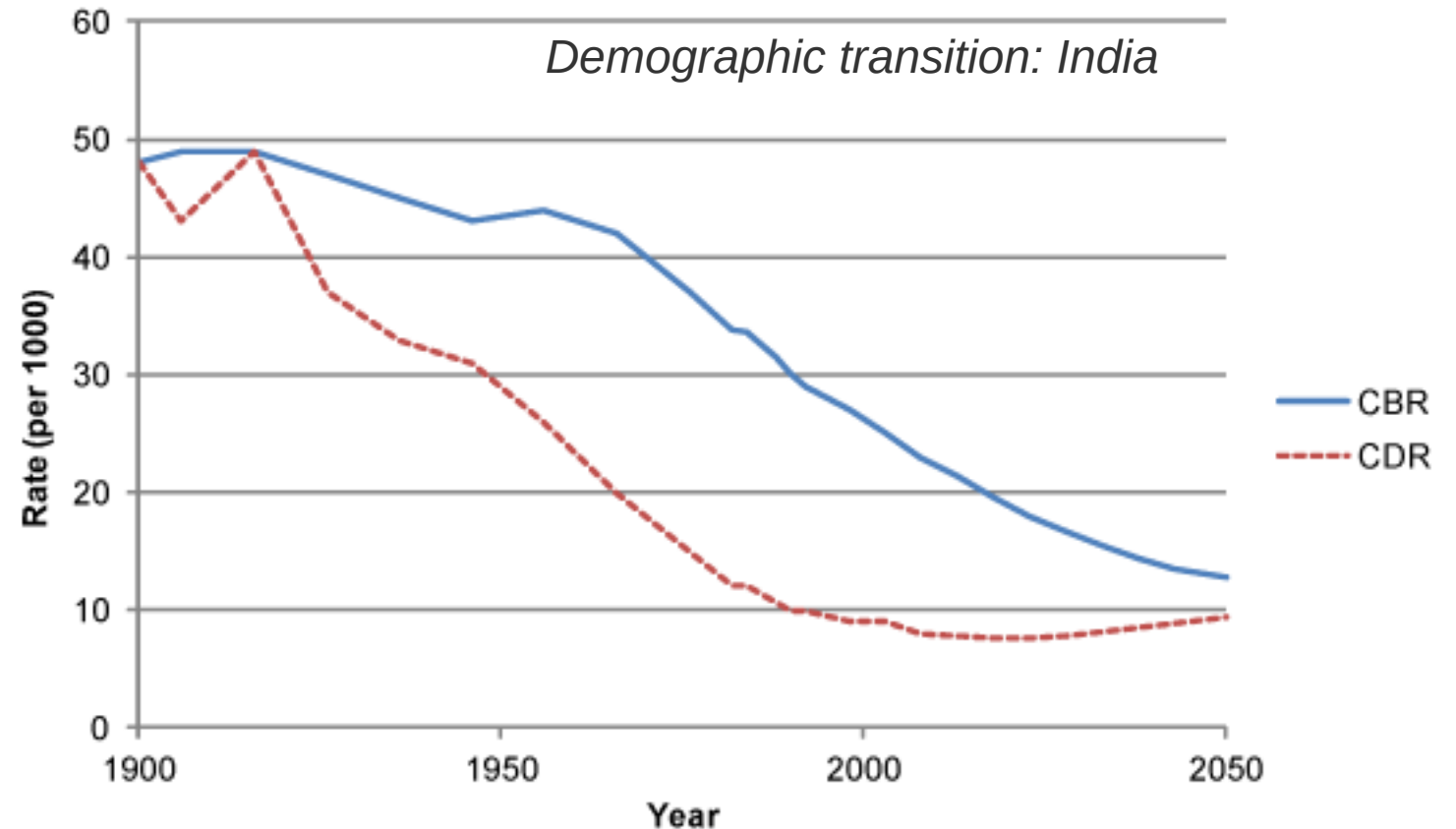
Negligible growth
rates.



INDIAN EXPERIENCE

- Timing
- Speed
- Growth rates
- Causes

despite occurring later, once mortality began to fall in the less developed countries, it fell far more quickly than during historical transitions.



MORTALITY DECLINE- CAUSES

Advances in public health

- ▢ Quarantine; Provision of clean water; Sanitation; vaccination campaigns

Improvements in nutrition-- reduces susceptibility to communicable diseases, especially in children

Widespread **acceptance of the germ theory of disease**

Medical advances

Other factors in developing countries

Socioeconomic development

- ▢ relationship between GDP per capita and life expectancy.

Medical technologies and public health interventions imported from elsewhere can rapidly reduce mortality.